

The barebones PageRank algorithm developed by Google's founders Sergey Brin and Larry Page in 1998 has received several major updates over the years for things like making search results more contextually relevant. As a result, the PageRank algorithm today is significantly more complex than the first rudimentary methodology laid down by Brin and Page. However, the basic idea remains the same. PageRank works by representing the entire internet as a graph. Mathematically, the graph's nodes represent the webpages, and an edge between two nodes in the graph represents a link between those two pages. Naturally, these graphs will be directed, i.e. the edges will have a direction specified because a page may link to, but may or may not be linked by, another page. The graph is then translated into a square matrix ($N \times N$) of gigantic dimensions since each row (column) represents links from (to) each page. For the entire internet, N easily approaches tens of billions.

The ranks of all the pages are computed simultaneously by finding the eigenvector of the matrix -- a specific "direction" in the matrix which, when tweaked, leaves the other "directions" unchanged. So for a pair of pages, the one with the higher rank will be displayed higher up in the search results.

The crucial step, and also the one most mathematically and computationally intensive, is the computation of the eigenvector. Fortunately, since it's not common for websites to randomly link to unrelated webpages, a huge majority of the matrix elements are 0, simplifying the calculations a little. It should also be no surprise that this step is the one that has been innovated upon the most. After Page and Brin's "power method", the revisions in the PageRank computations have included the accelerated power method, aggregation/disaggregation methods, Krylov methods and Schwarz methods, spanning a total of seven years starting from 1999.

Building a better mousetrap

Starting with 2011, each year has seen one major headline update to the software, respectively codenamed Google Panda, Google Penguin, Google Hummingbird, and Google Pigeon.

Panda focused on downgrading "low-quality" or "thin" websites, and as a result news sites and social media websites gained prominence in Google's search results. Penguin was targeted at websites that used search engine optimisation (SEO) techniques that amounted to spamming or link manipulation to make the website seem more important than its content deserves to be.